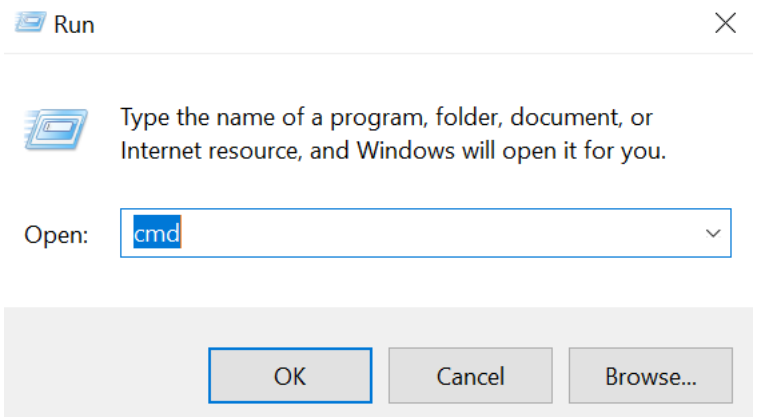


WINDOWS

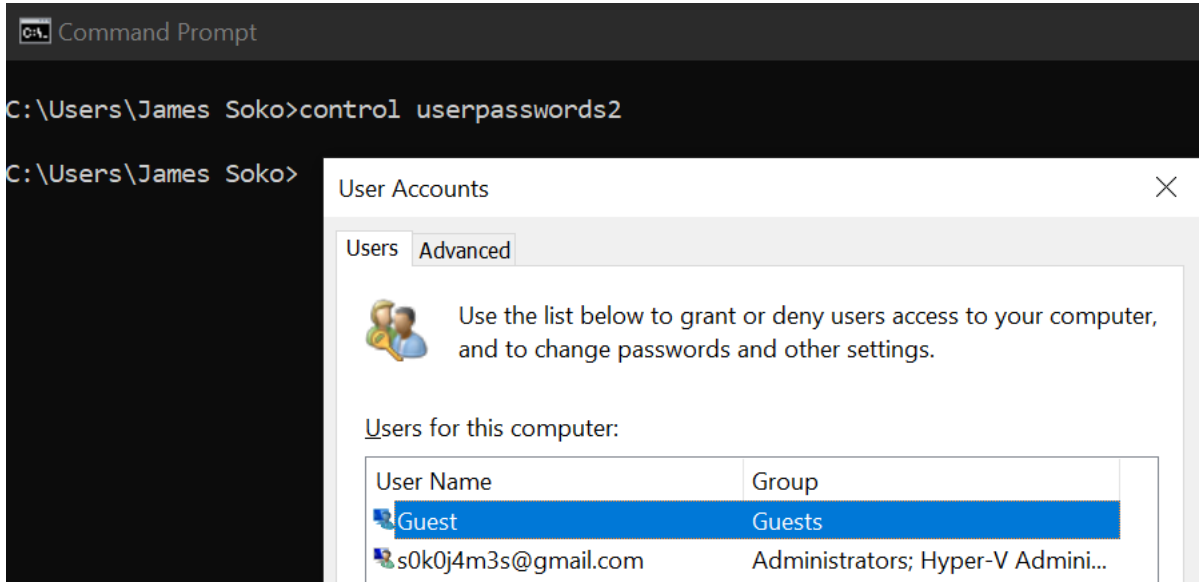
Cracking a User Password From The SYSTEM & SAM File Using John The Ripper

SETTING UP A PASSWORD

1. Press + R on the keyboard to access the Run dialog and type CMD to open **Command Prompt.**

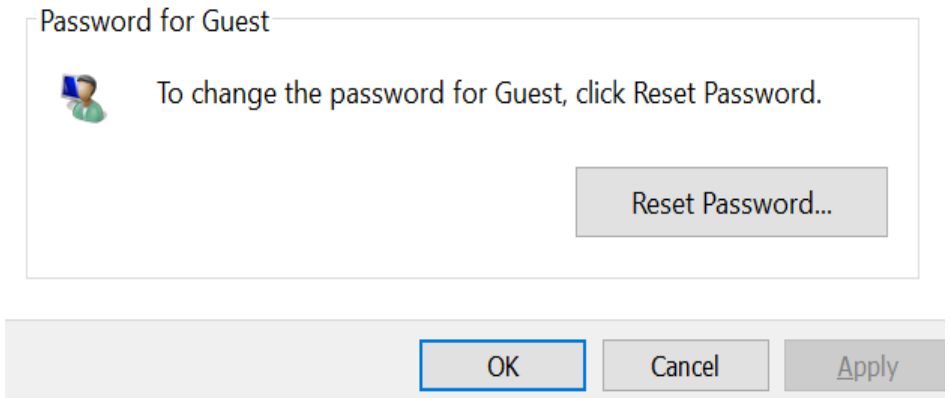


2. Type the command below:

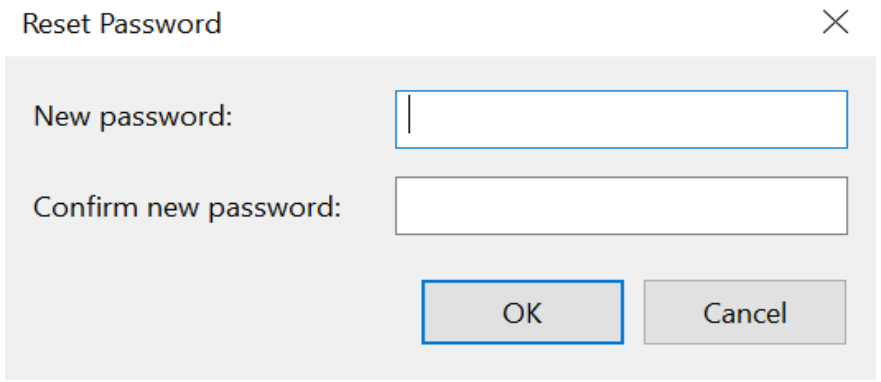


SETTING UP A PASSWORD (CONTINUES…)

3. Click on **Reset Password…**

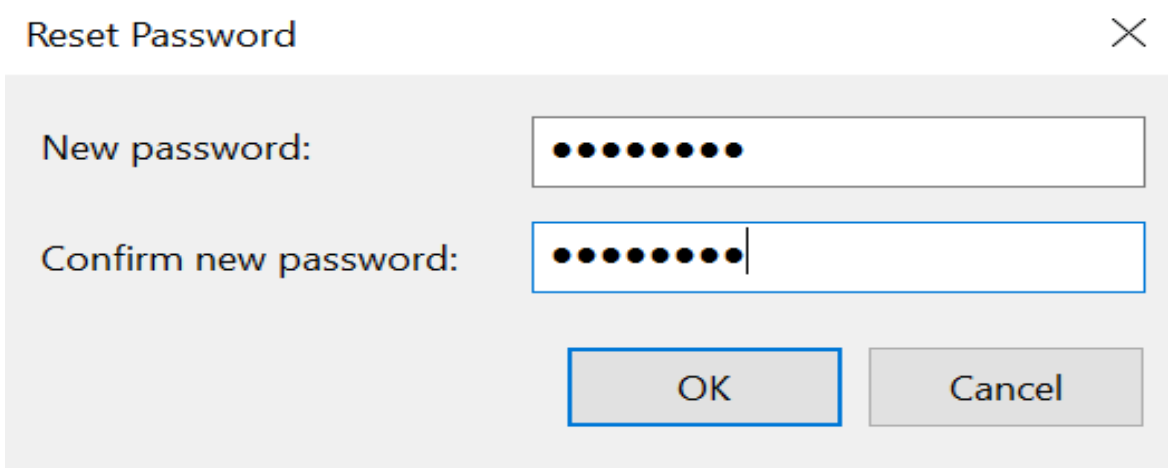


4. We need to set up a password for our lab.



SETTING UP A PASSWORD (CONTINUES…)

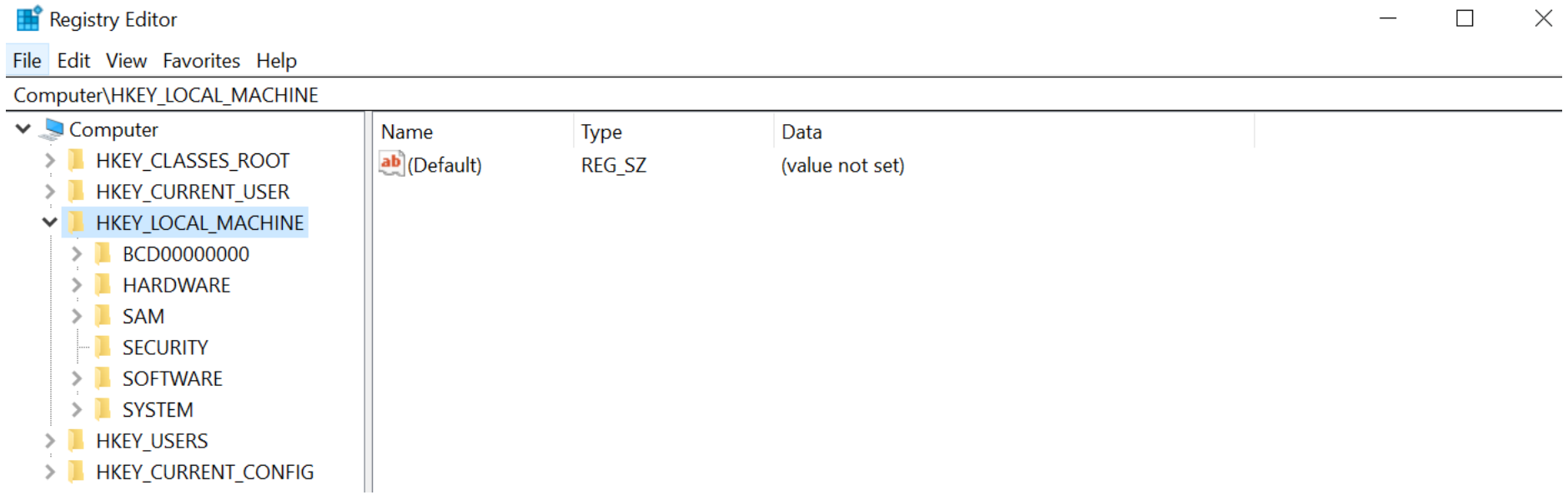
5. Enter the password you want to set up, for the lab will use the password: **thisisit**



6. Click on **OK**

ACCESSING THE SYSTEM AND SAM FILES

- Before we start the lab, we need to know where these files are stored.
- Search for **Registry Editor** on your Windows Search bar.



ACCESSING THE SYSTEM AND SAM FILES (CONTINUES…)

- Notice our **SYSTEM** and **SAM** files are under **HKEY_LOCAL_MACHINE**.
- The initials for HKEY_LOCAL_MACHINE is **HKLM** because H is for Hive, so its considered as a separate word.
- The **hklm** will be used as our command to **save** the SYSTEM and SAM files because you can't copy and paste these files directly.
- To make our work easy, let's create a folder on our Desktop, let's just call it Windows.



ACCESSING THE SYSTEM AND SAM FILES (CONTINUES…)

1. Open Command Prompt as an Administrator inside the current created Folder.



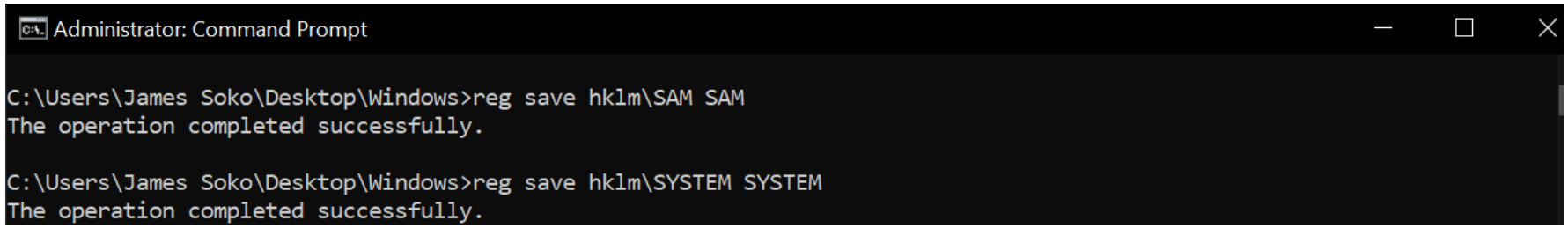
2. **NOTE:** Registry Editor → HKEY_LOCAL_MACHINE → SYSTEM/SAM is the path, think of it has our command.
3. Commands

a) reg save hklm\SAM SAM

b) reg save hklm\SYSTEM SYSTEM

ACCESSING THE SYSTEM AND SAM FILES (CONTINUES…)

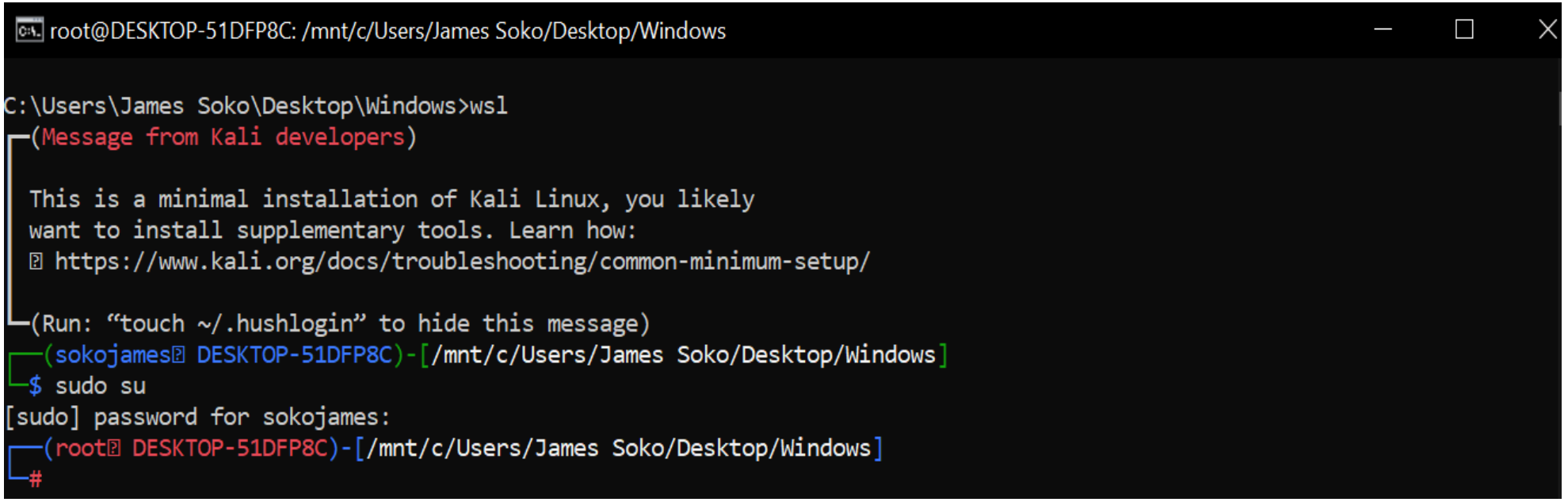
4. **NOTE:** The **reg save hklm\SAM SAM** indicates that the first SAM is the actual SAM file and the second SAM is the new file name used to save it, you can use your own default names e.g., reg save hklm\SAM samfile. This also applies to the **SYSTEM** file.





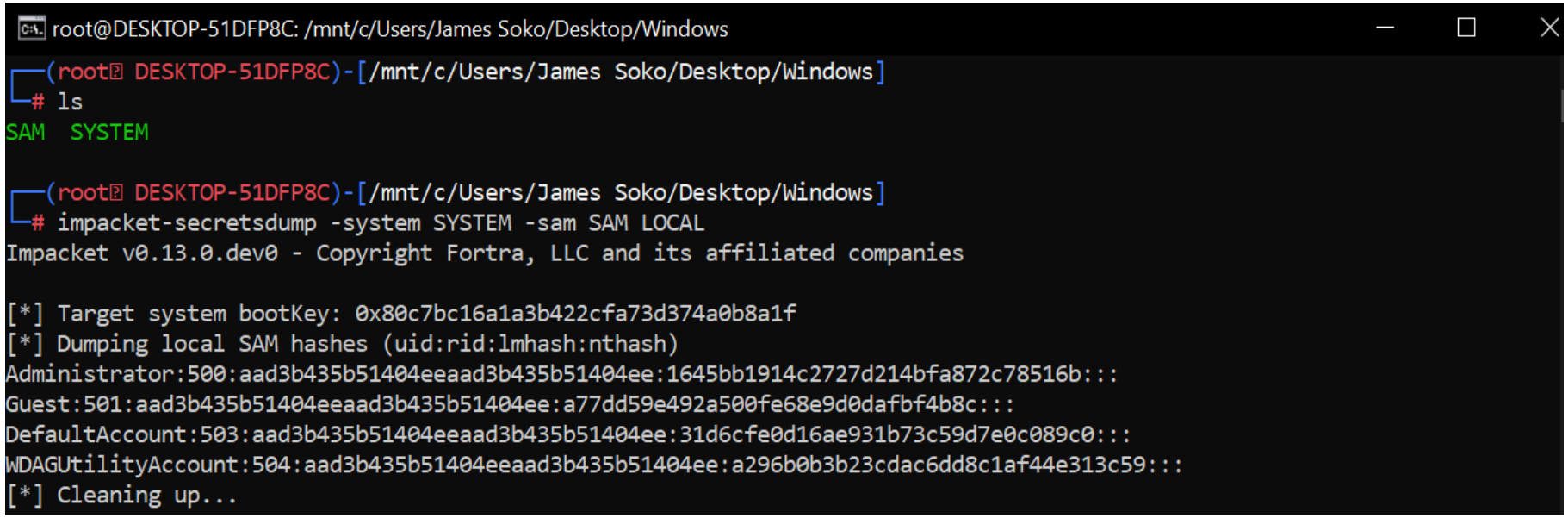
CRACKING THE WINDOWS USER PASSWORD FROM THE SYSTEM AND SAM FILE USING JOHN THE RIPPER

- To achieve this, you need a Linux system, in my case, I have Windows Subsystem for Linux (WSL) installed.



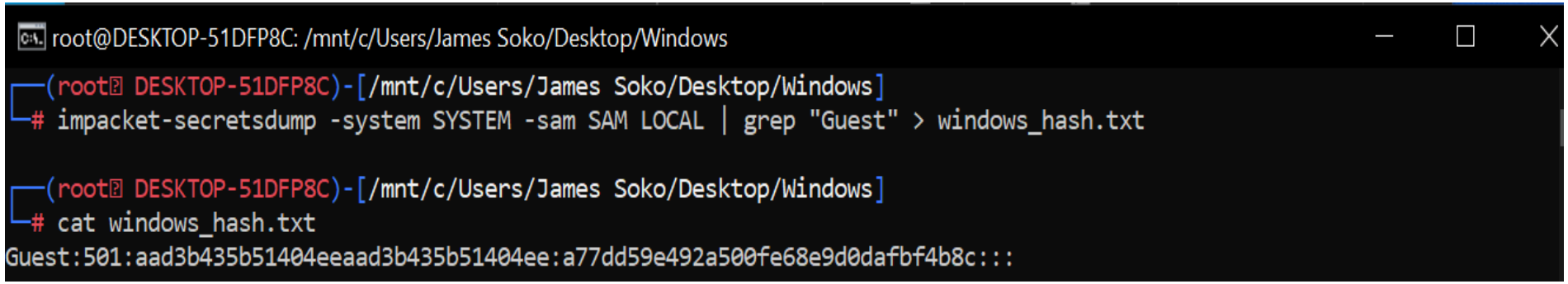
CRACKING THE WINDOWS USER PASSWORD FROM THE SYSTEM AND SAM FILE USING JOHN THE RIPPER (CONTINUES…)

1. Let's extract the hashes from the SYSTEM and SAM files and then merge them using **impacket-secretsdump**.



CRACKING THE WINDOWS USER PASSWORD FROM THE SYSTEM AND SAM FILE USING JOHN THE RIPPER (CONTINUES…)

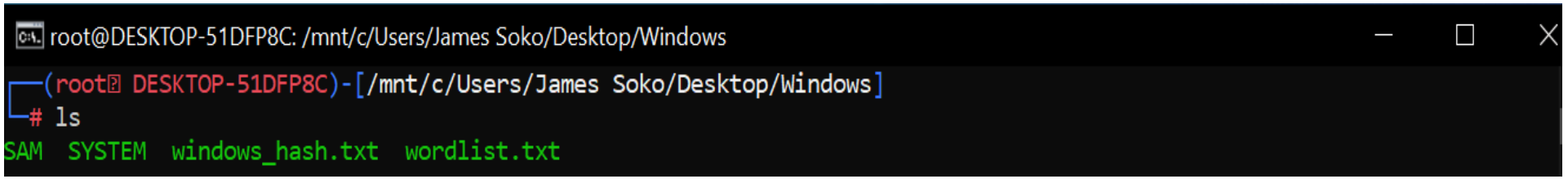
2. **NOTE:** Our target account in this case is the **Guest** account, based on your target, get the hash file for that account.



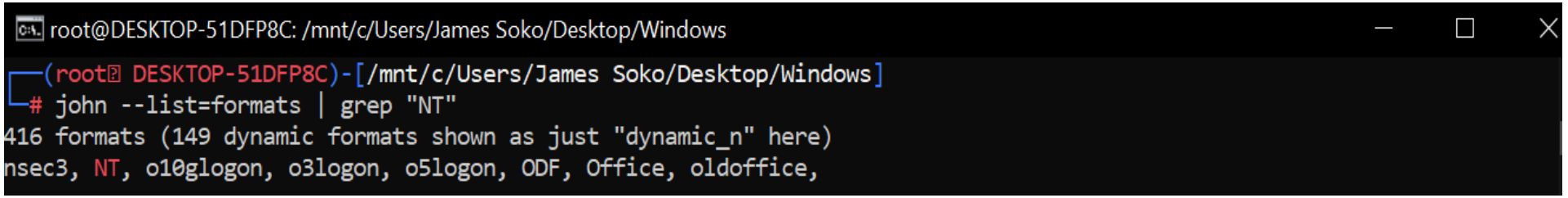
3. At this point, we need a wordlist that will be used to crack the password using **John The Ripper**.

CRACKING THE WINDOWS USER PASSWORD FROM THE SYSTEM AND SAM FILE USING JOHN THE RIPPER (CONTINUES…)

4. Place the wordlist in the same directory as your files.

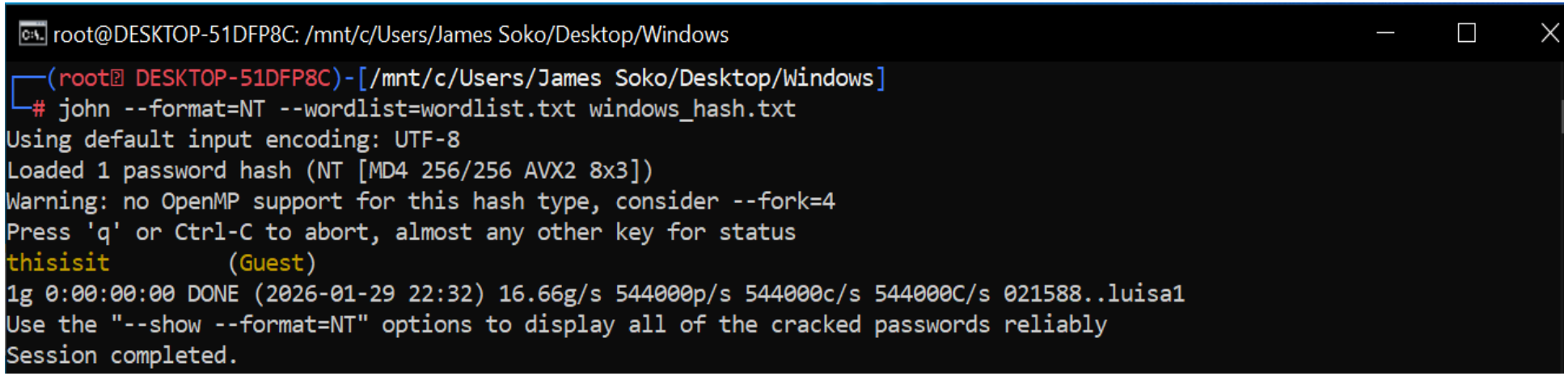


5. This Windows computer is running on Windows 10 and it happens to be part of the New Technology (NT), so with john, we need to specify the format.



CRACKING THE WINDOWS USER PASSWORD FROM THE SYSTEM AND SAM FILE USING JOHN THE RIPPER (CONTINUES…)

6. Finally, we crack the password:



- **RESOURSES: YouTube** Tutorial and **John The Ripper**



OPERATING SYSTEMS SECURITY

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